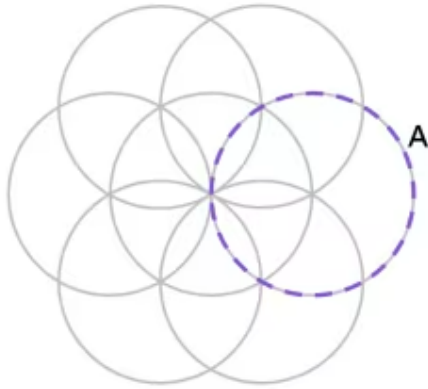


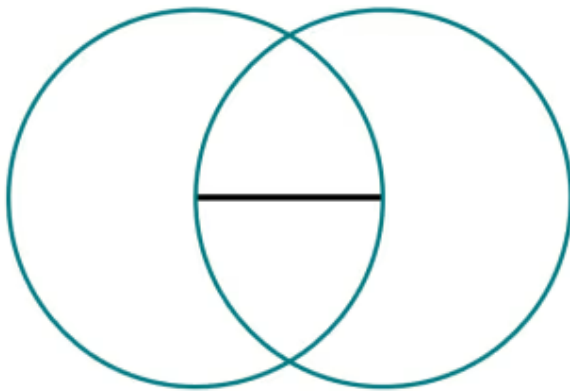


Problem solving with constructions

- 1 In this six-petal flower construction, at how many points does circle A intersect with at least one other circle?



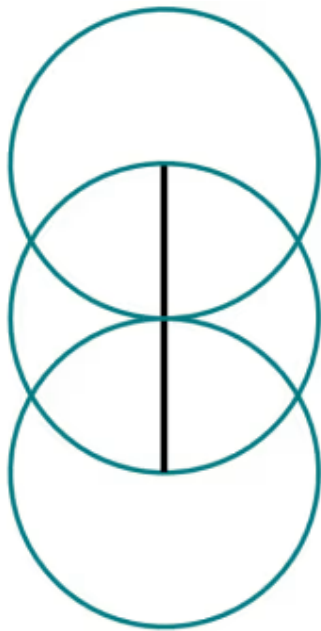
- 2 Which of these polygons can be formed by joining up combinations of intersections between any line segment or circle? (Tick **3** correct answers)



- ☐ equilateral triangle
- ☐ trapezium
- ☐ scalene triangle
- ☐ rhombus
- ☐ square



- 3 Which of these polygons can be formed by joining up combinations of intersections between any line segment or circle? (Tick 4 correct answers)



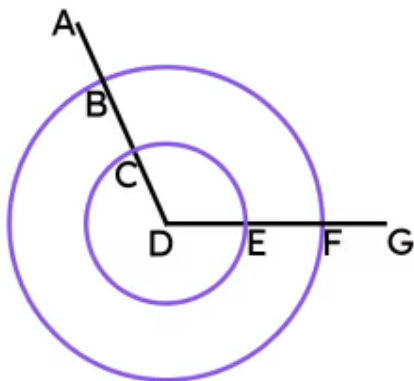
- ☐ kite
- ☐ parallelogram
- ☐ square
- ☐ trapezium
- ☐ pentagon

- 4 Which of these statements is true about the construction of a perpendicular bisector? (Tick 3 correct answers)

- ☐ It is formed from the intersections of two congruent circles.
- ☐ It is formed from intersections of different pairs of congruent circles.
- ☐ It can be formed from the intersections of any two circles.
- ☐ At least one circle must have its centre at the midpoint of a line segment.
- ☐ All circles must have their centres at the endpoints of a line segment.



- 5 Which two pairs of points need to be joined with a line segment to find the fourth point in the kite? (Tick 2 correct answers)



- ☐ AE
☐ BE
☐ CG
☐ CF
☐ BC

- 6 This "crescent moon" logo can be formed using the construction of an equilateral triangle that uses two congruent circles of radius 9 cm. What is the perimeter of this logo (in cm, rounded to 1 dp)?

